

Corrected reproduction of the LEC climatology figures

A new 16-figure report based on the validated LorenzCycleToolKit 2.0.0 rerun for Southwestern Atlantic cyclones

Corrected cyclones	3,820
Primary cyclone-phase rows	15,280
Toolkit commit	d38cda7e37d8
Correction commit	d07707767c29
Figures	16 PNG + 16 PDF

Purpose

This document recreates the scientific roles of Figures 1-16 in the published LEC climatology article while replacing every LEC-dependent value with the corrected 3,820-case rerun. Published legacy figures remain untouched. Track-only and conceptual panels are regenerated from their frozen source data and definitions.

Scope decisions

The main panels use one primary incipient, intensification, mature and decay record per cyclone. Secondary lifecycle episodes remain preserved upstream but are outside this four-phase reproduction. EOFs use the published 24-term correlation-matrix definition. Intense groups retain the published 90th-percentile vorticity criterion and four-cluster structure, with deterministic initialization for reproducibility.

Updated numerical overview

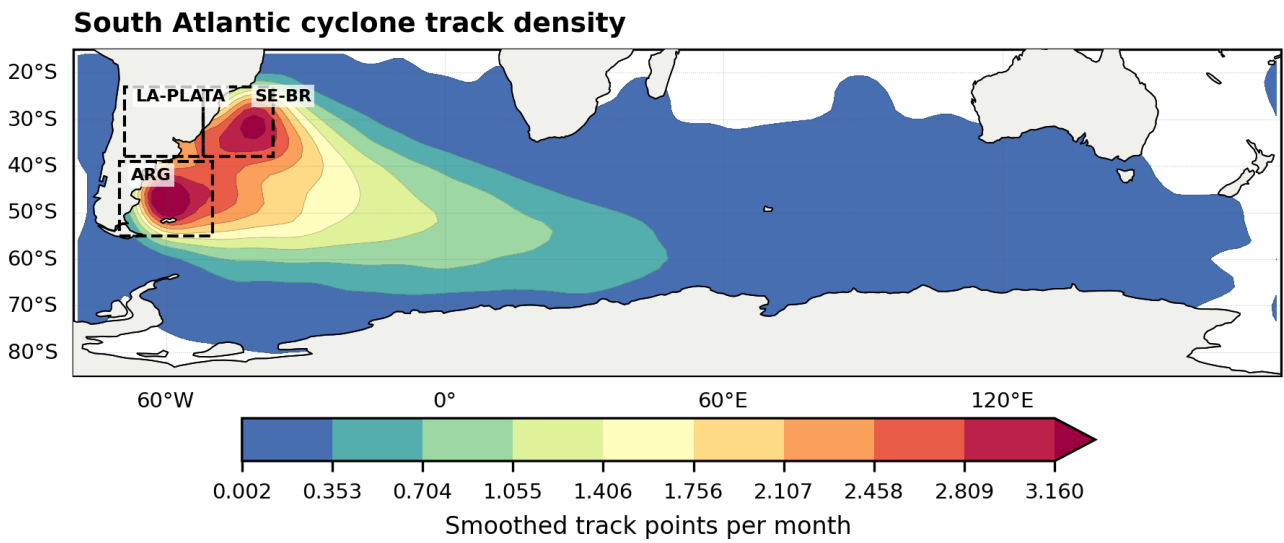
Phase	Ca mean	Ck mean	Ce mean	EOF1 variance
incipient	1.92	0.33	3.80	20.58%
intensification	4.12	-0.30	6.23	27.55%
mature	3.67	-2.43	4.55	26.30%
decay	2.16	-2.10	1.98	29.17%

Intense-cyclone groups

The pointwise vorticity threshold is 8.293; 1,486 corrected cyclones enter the four-group analysis.

Cluster	n	Mean maximum vorticity	Median maximum vorticity
1	745	9.82	9.60
2	421	10.84	10.61
3	193	10.14	9.86
4	127	11.04	11.01

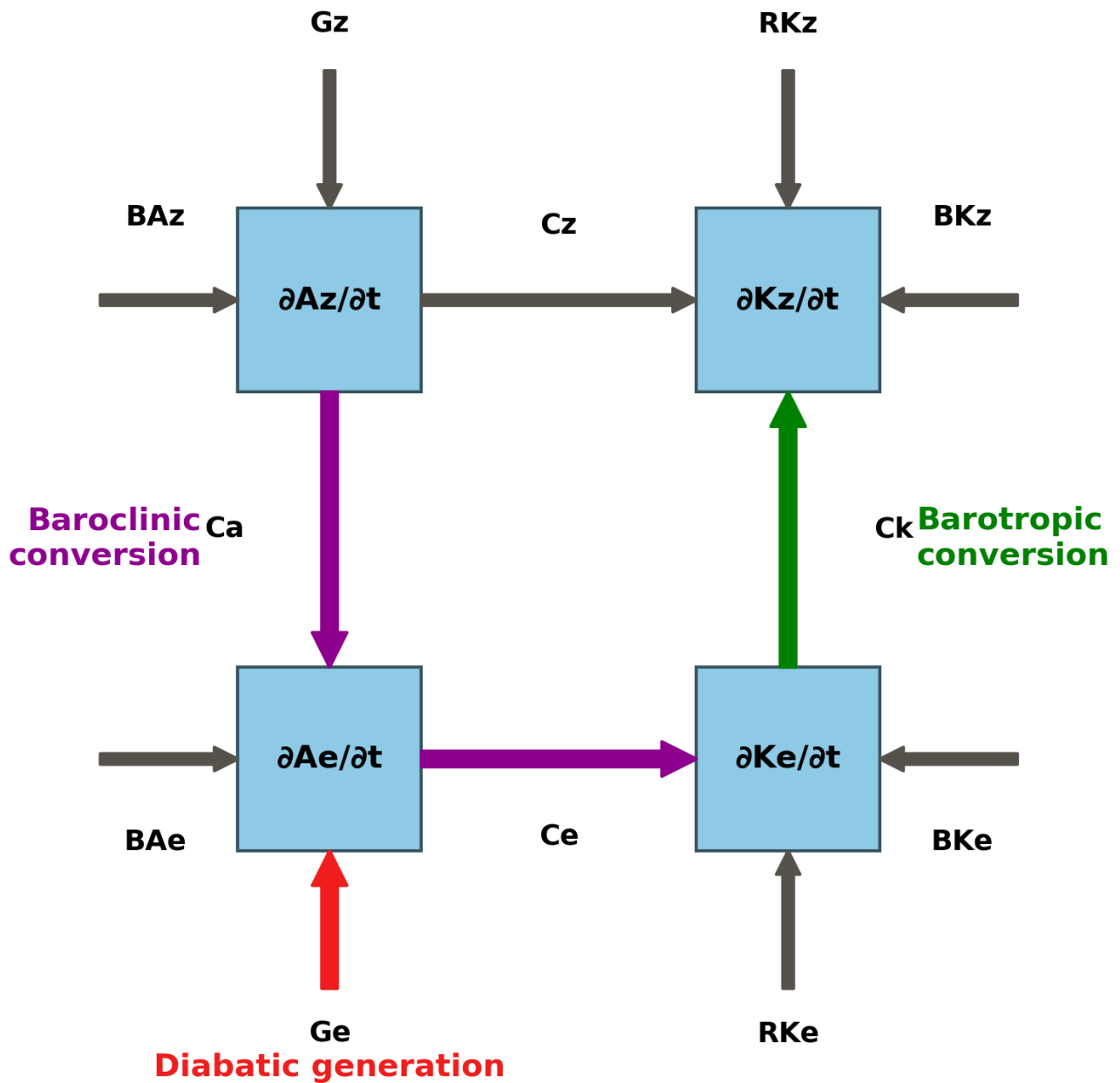
Figure 1



Track density of the South Atlantic cyclone database and the three coastal genesis regions used in the analysis.

Figure 2

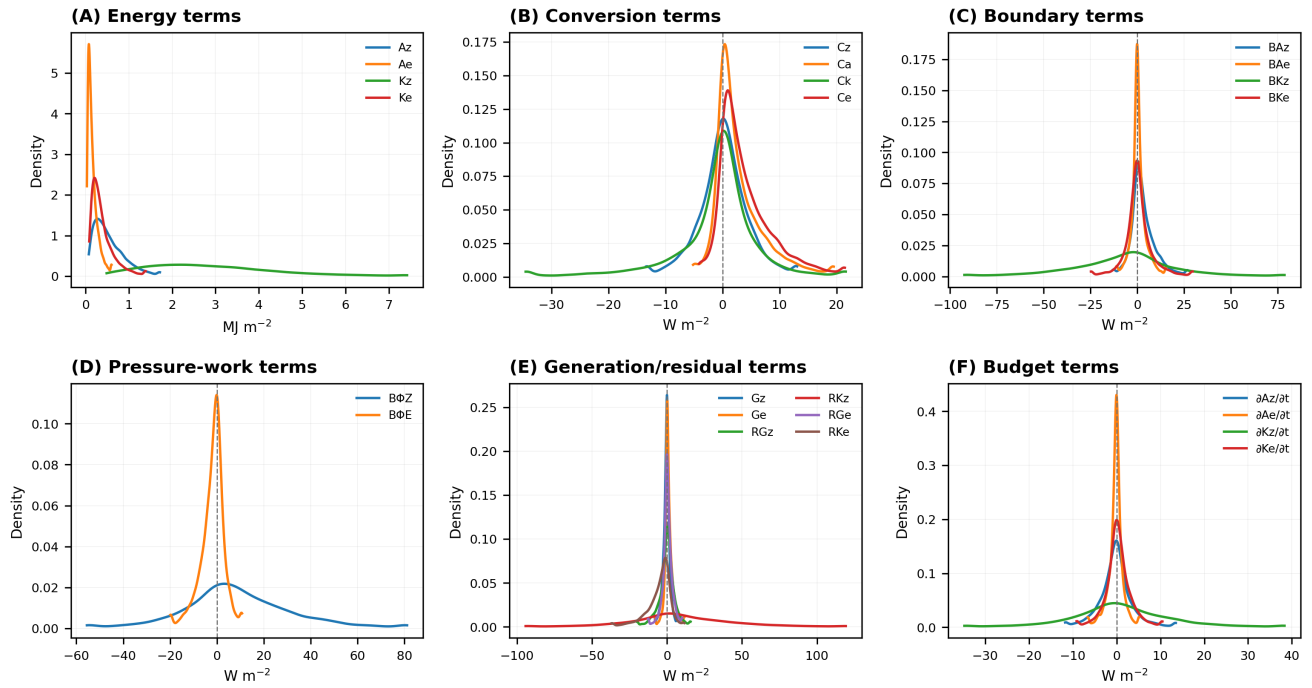
Lorenz Energy Cycle reference diagram



Reference Lorenz Energy Cycle diagram and the principal physical pathways discussed in the article.

Figure 3

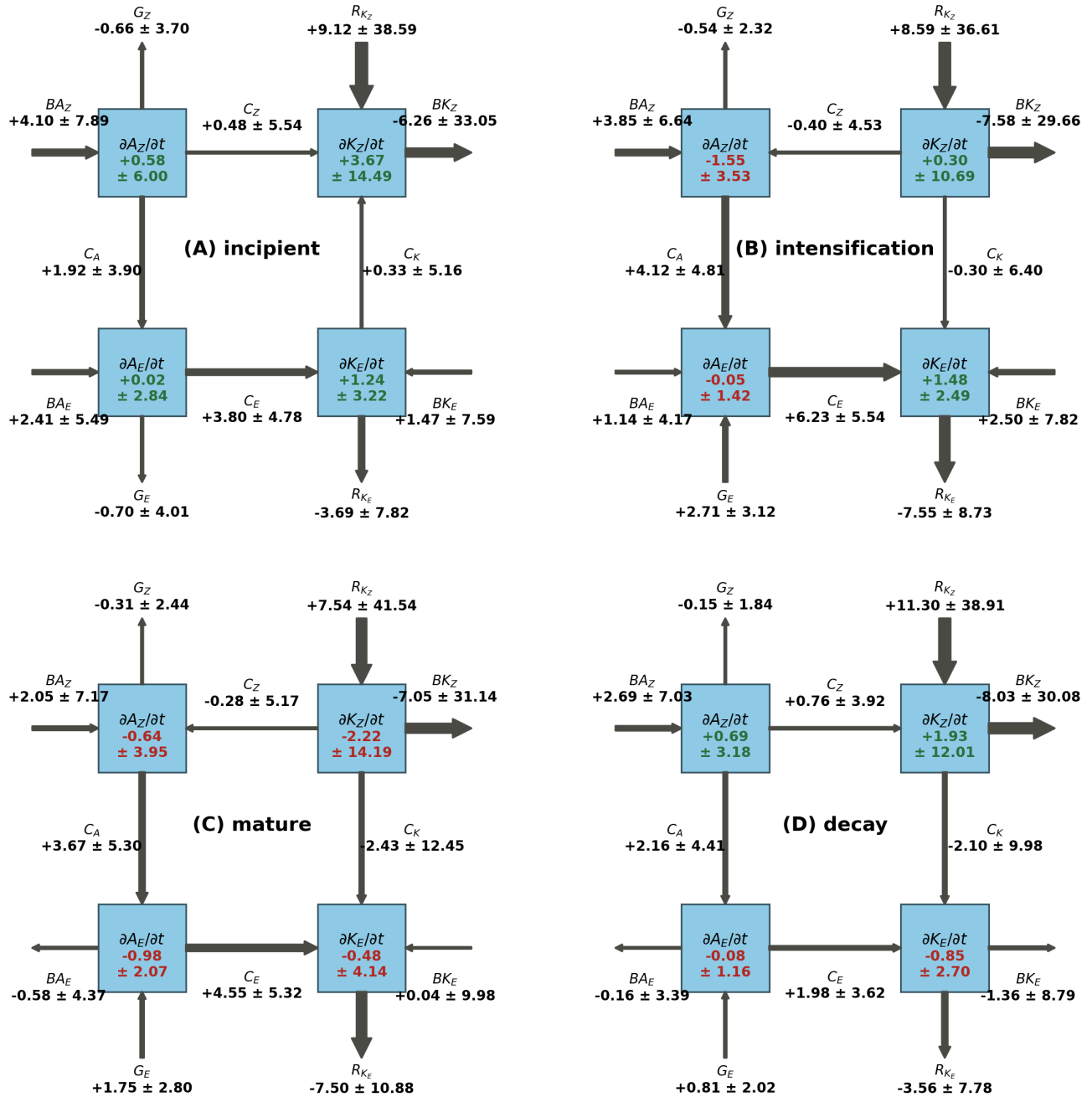
Corrected LEC term distributions - primary lifecycle phases



Corrected probability-density functions of energy, conversion, boundary, pressure-work, generation/residual and budget terms.

Figure 4

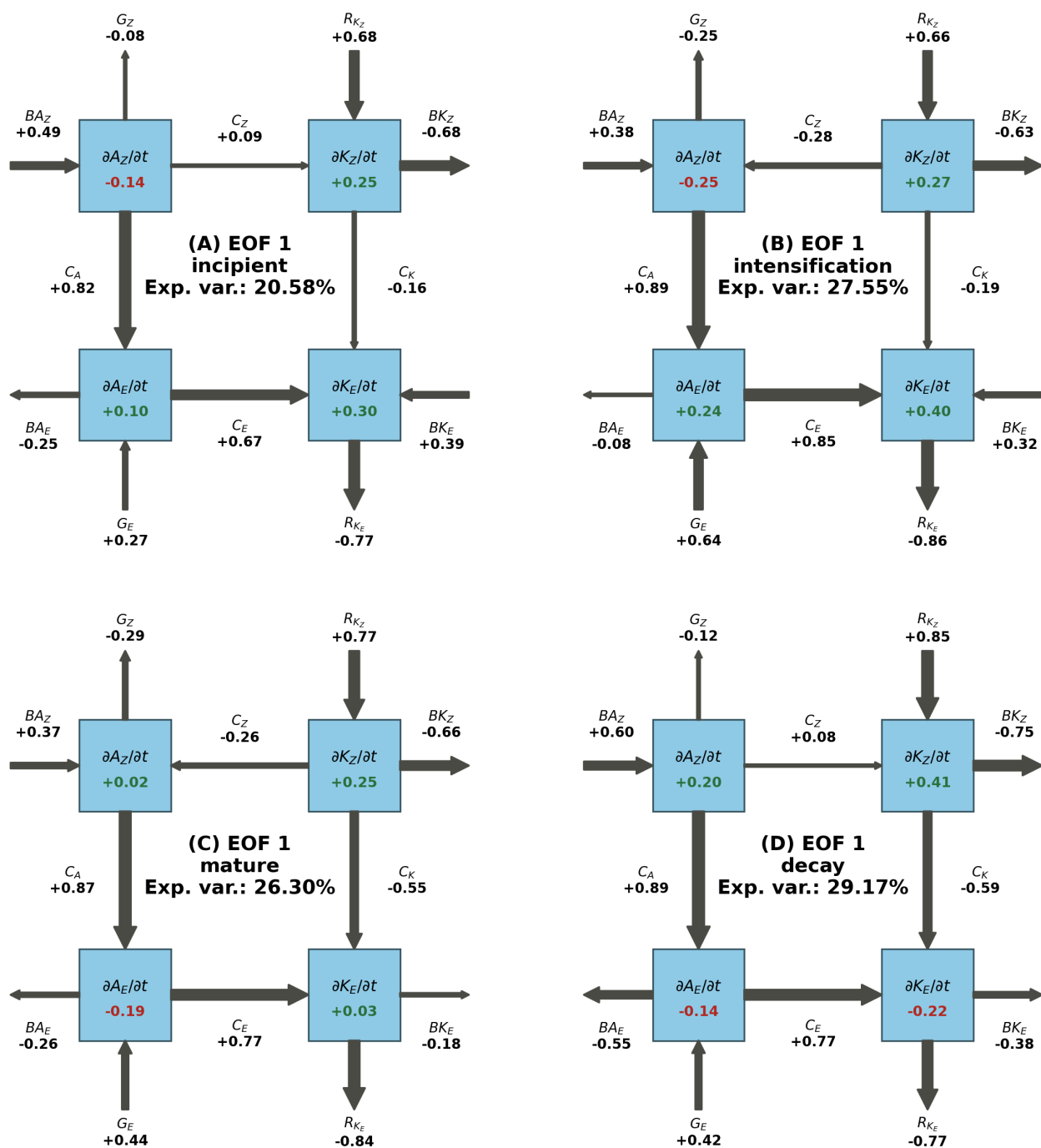
Corrected Lorenz Energy Cycle by lifecycle phase mean $\pm$ sample standard deviation



Corrected mean Lorenz Energy Cycle by primary lifecycle phase; labels show mean plus or minus one sample standard deviation.

Figure 5

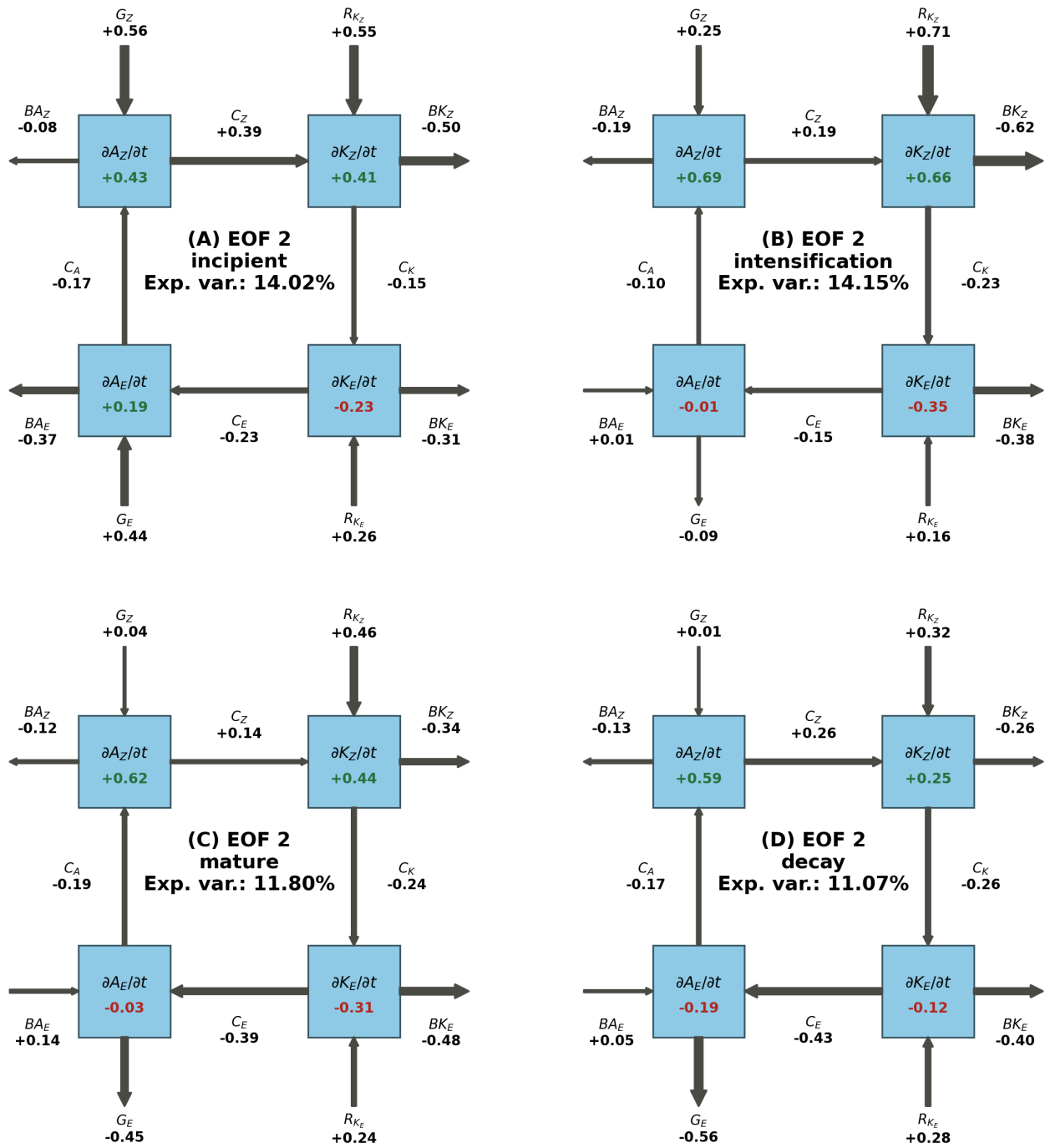
Corrected EOF 1 loadings by lifecycle phase



Corrected EOF 1 loadings on the Lorenz Energy Cycle by lifecycle phase.

Figure 6

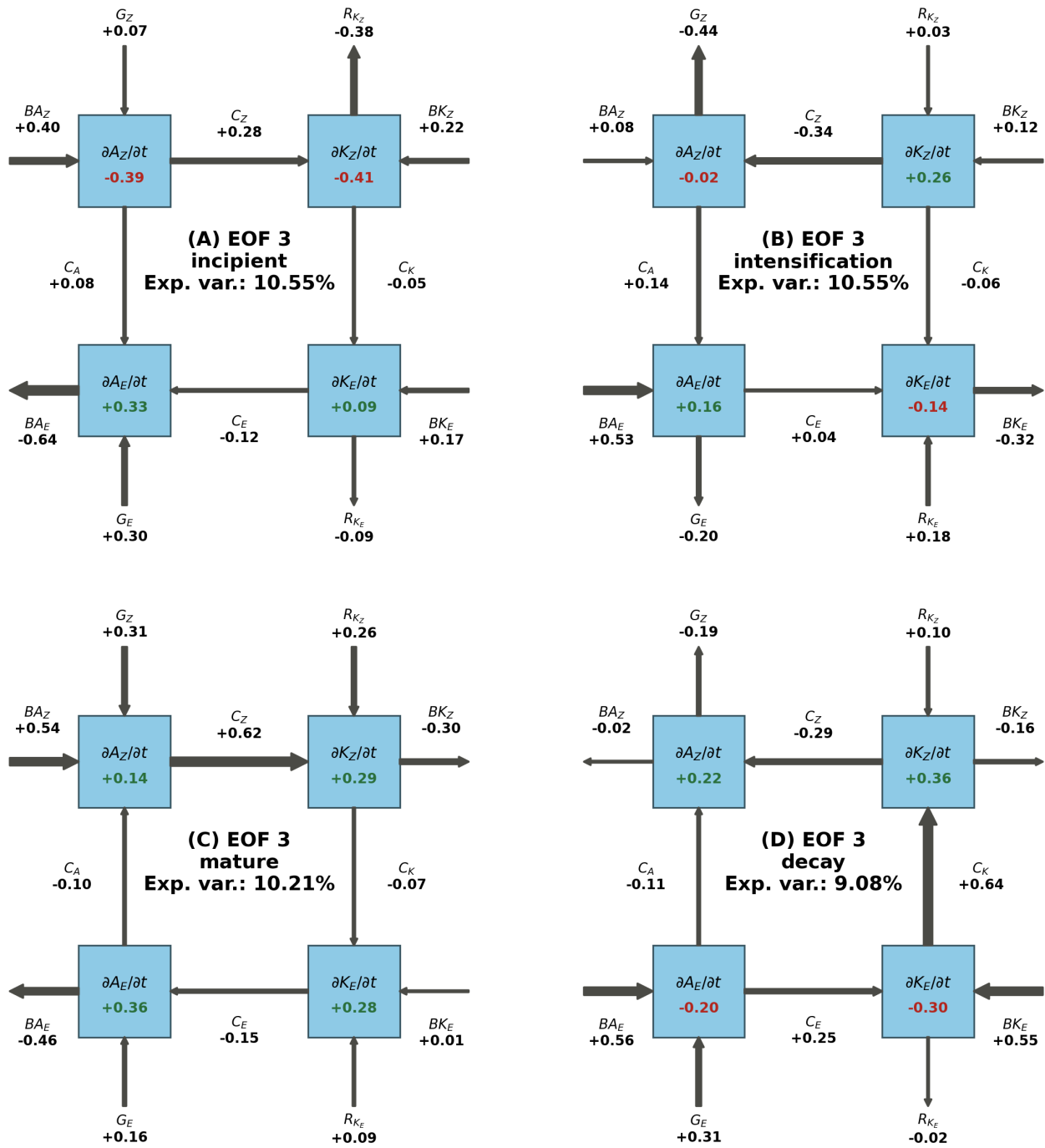
Corrected EOF 2 loadings by lifecycle phase



Corrected EOF 2 loadings on the Lorenz Energy Cycle by lifecycle phase.

Figure 7

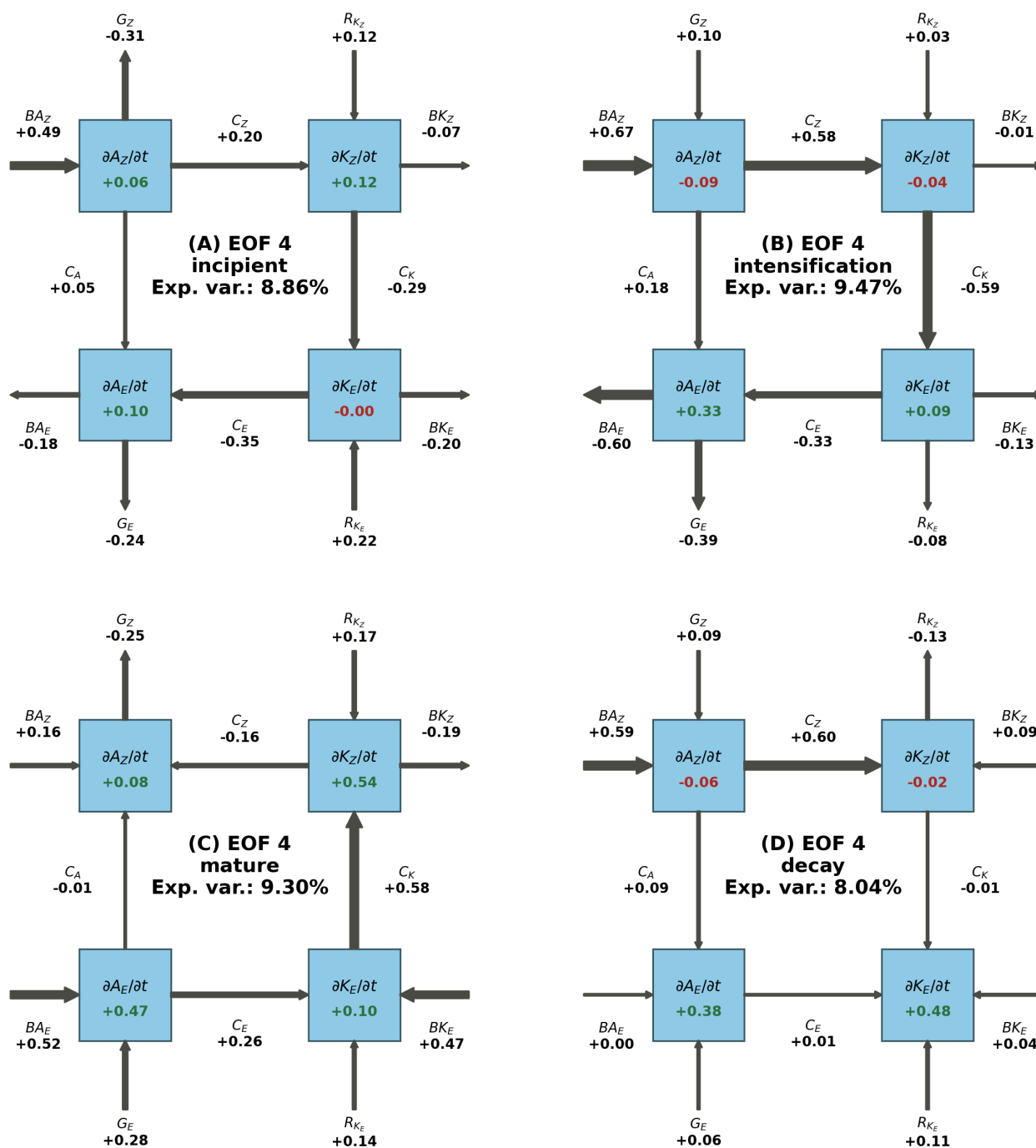
Corrected EOF 3 loadings by lifecycle phase



Corrected EOF 3 loadings on the Lorenz Energy Cycle by lifecycle phase.

Figure 8

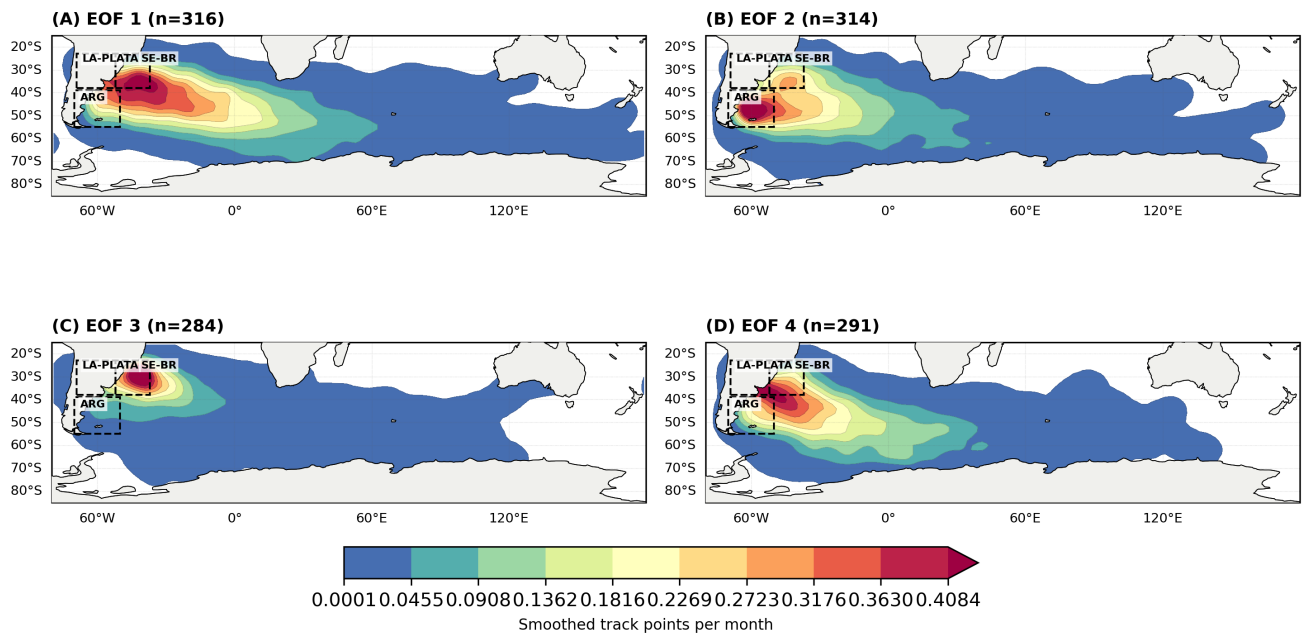
Corrected EOF 4 loadings by lifecycle phase



Corrected EOF 4 loadings on the Lorenz Energy Cycle by lifecycle phase.

Figure 9

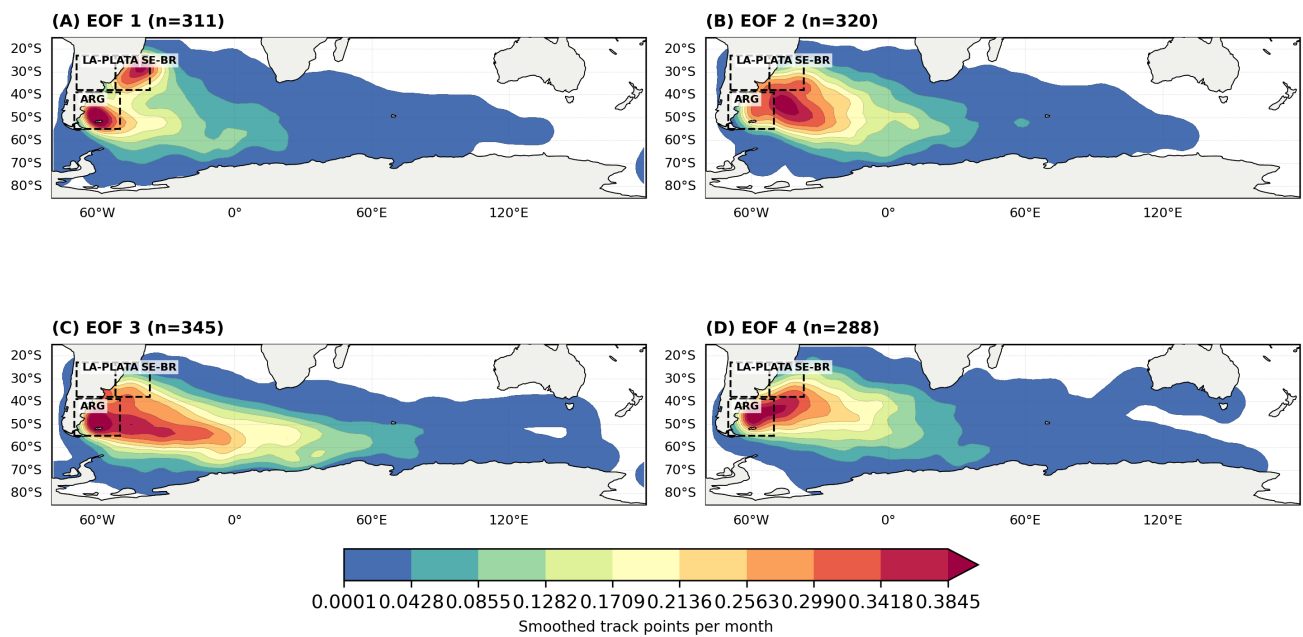
Track density - positive total-lifecycle PC extremes



Track densities for cyclones in the positive (upper-decile) extremes of total-lifecycle PCs 1-4.

Figure 10

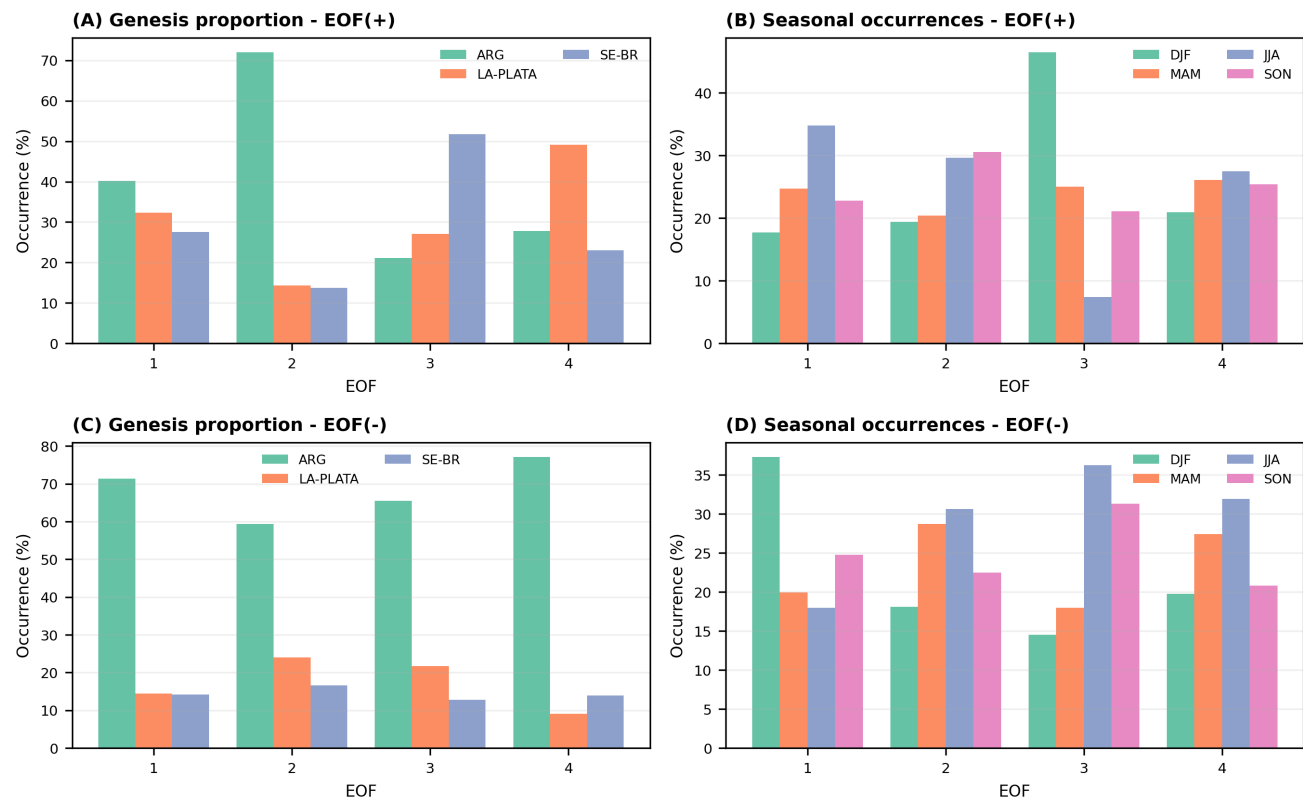
Track density - negative total-lifecycle PC extremes



Track densities for cyclones in the negative (lower-decile) extremes of total-lifecycle PCs 1-4.

Figure 11

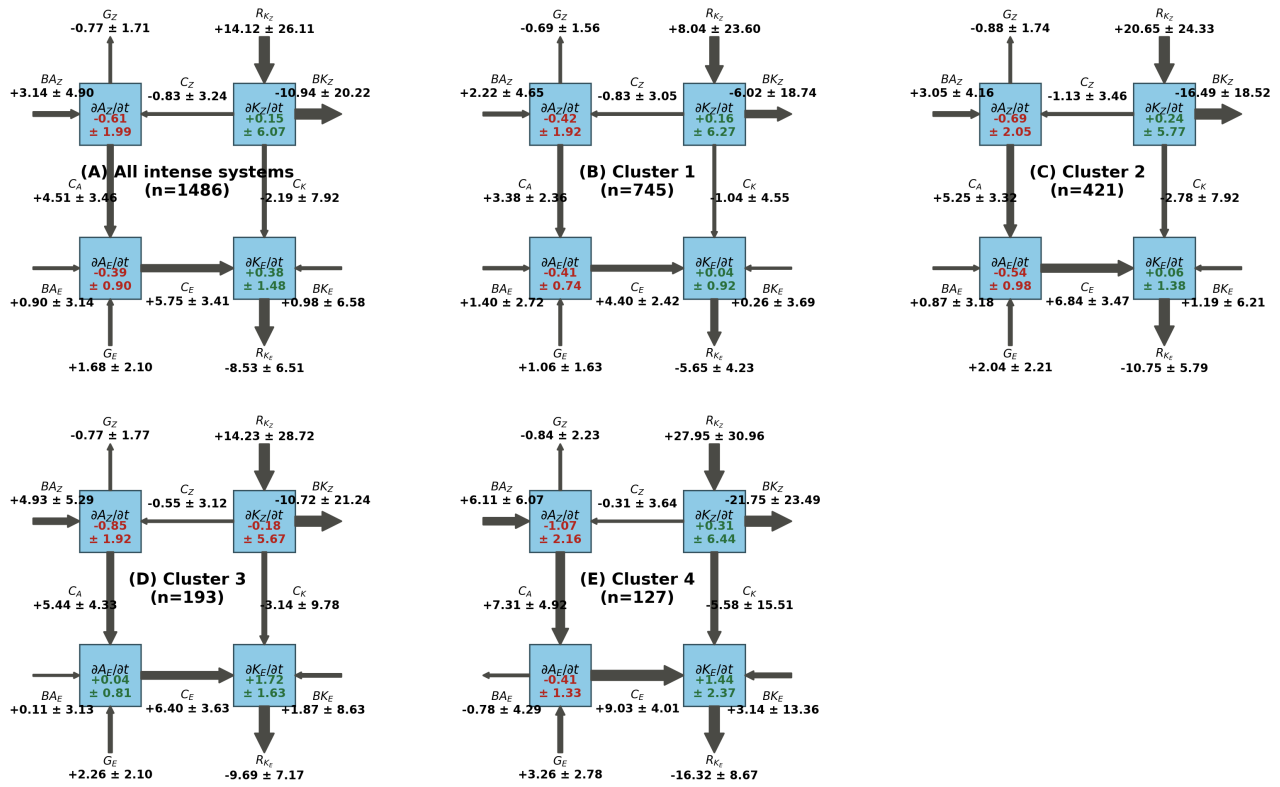
Corrected total-lifecycle EOF extremes



Genesis-region and seasonal composition of the positive and negative PC extremes.

Figure 12

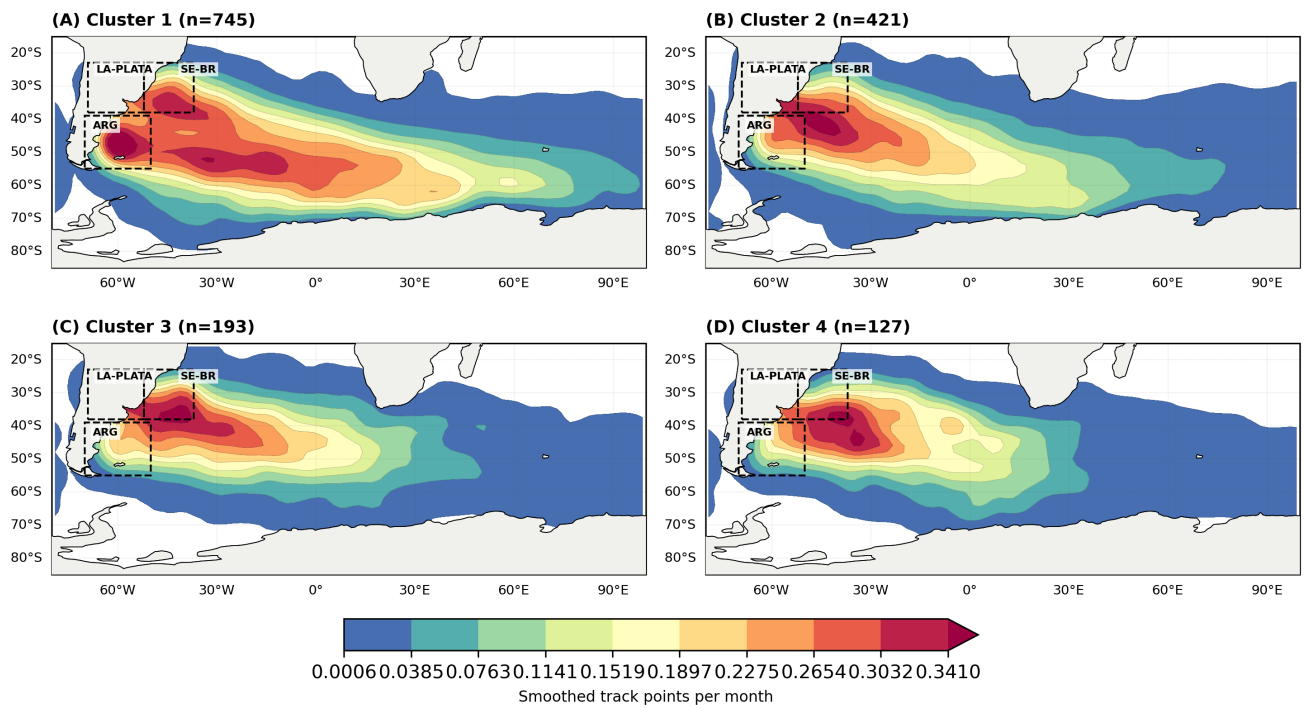
Corrected LEC groups among intense cyclones



Corrected mean Lorenz Energy Cycle for the intense-cyclone subset and its four deterministic K-means groups.

Figure 13

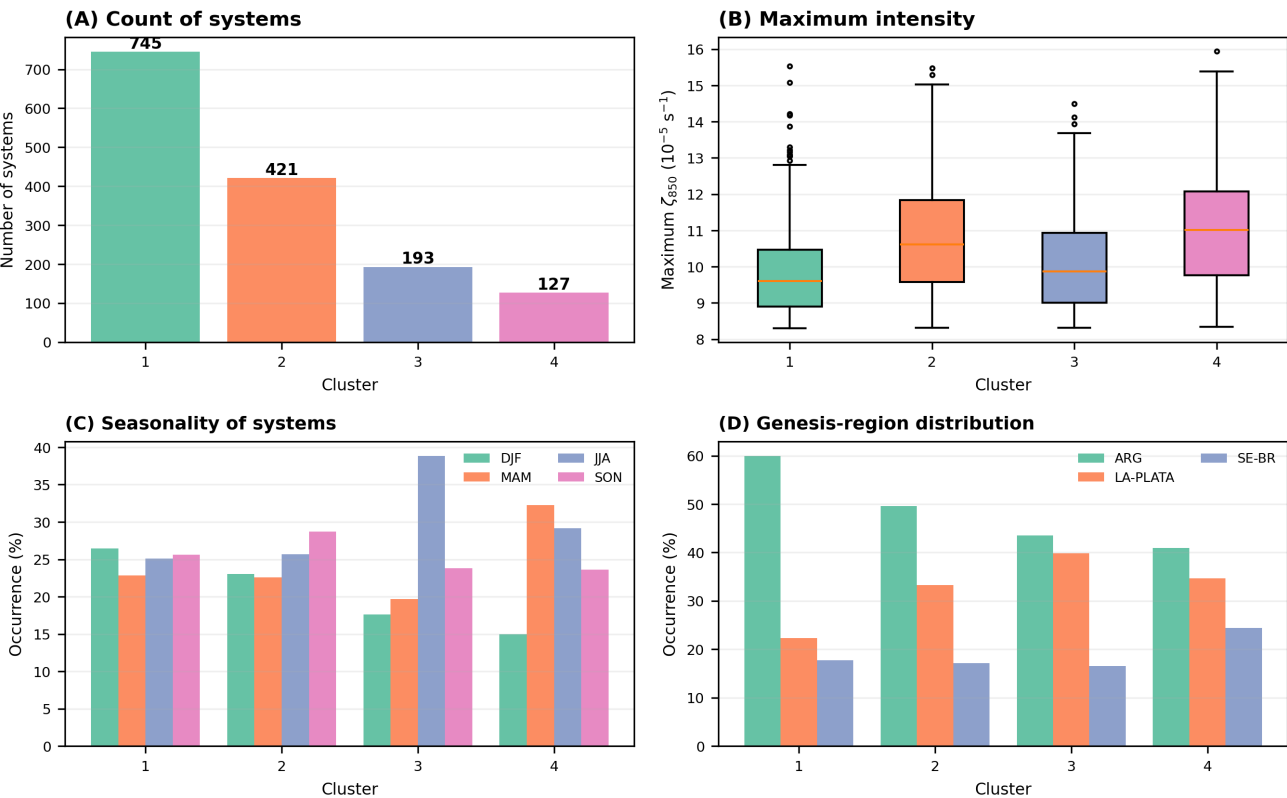
Track density of corrected intense-cyclone groups



Track densities of the four corrected intense-cyclone energy groups.

Figure 14

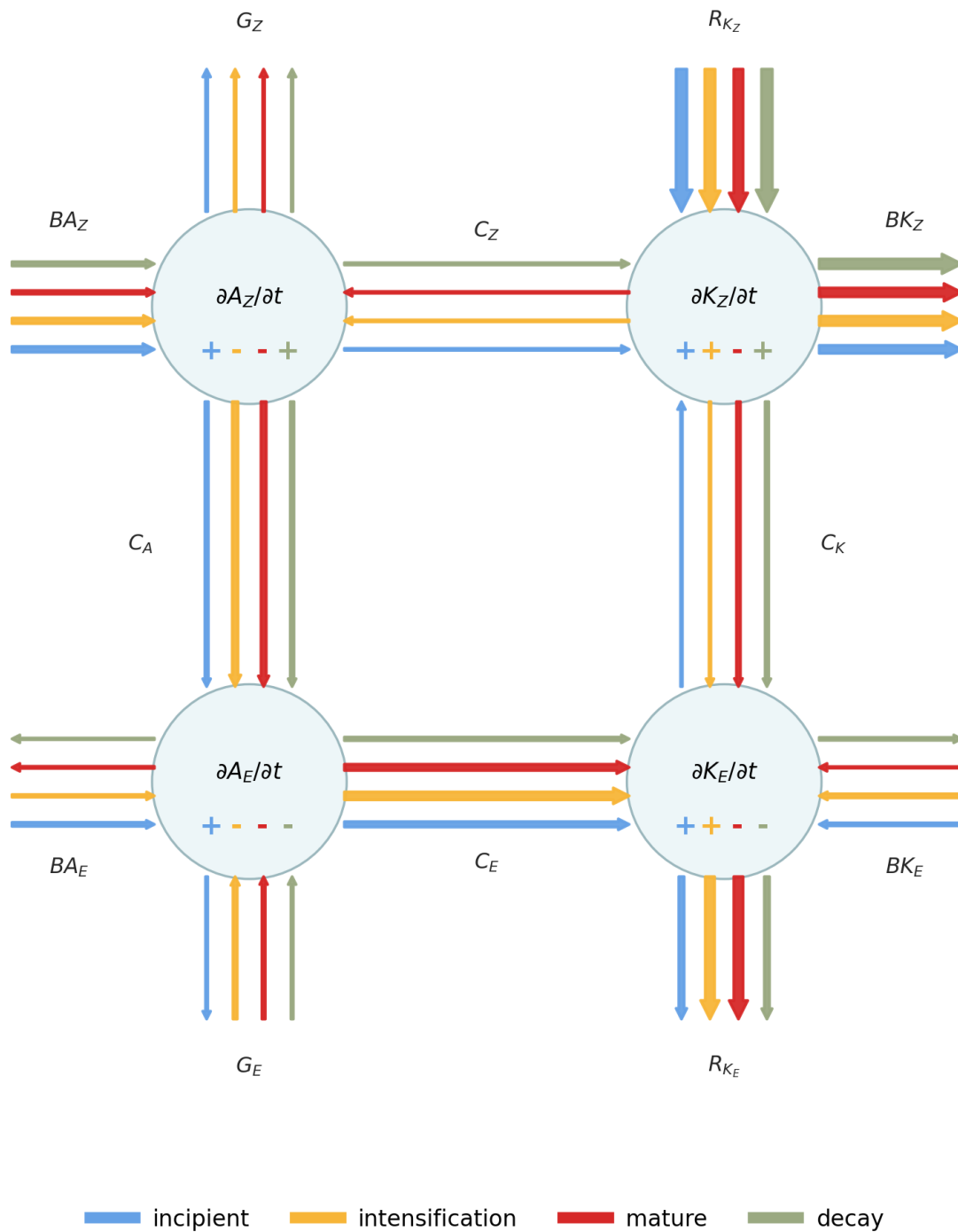
Corrected intense-cyclone group characteristics



Size, maximum intensity, seasonality and genesis-region composition of the corrected intense-cyclone groups.

Figure 15

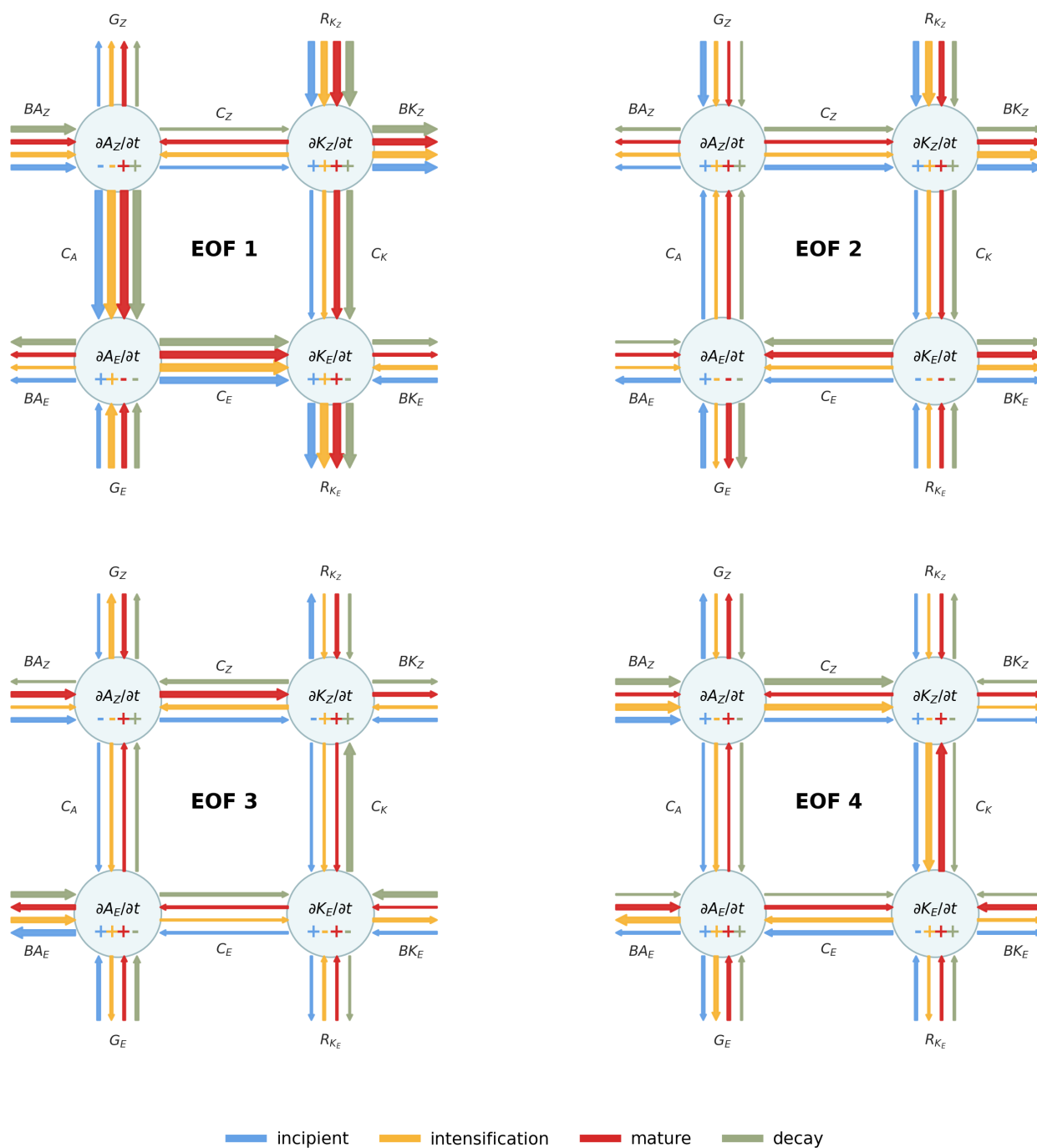
Corrected phase-mean synthesis



Synthesis of the corrected mean Lorenz Energy Cycle across the four primary lifecycle phases.

Figure 16

Corrected EOF synthesis by lifecycle phase



Synthesis of corrected EOFs 1-4 across the four primary lifecycle phases.

Reproducibility record

Every figure has a PNG and PDF checksum in `results/lec_climatology_corrected/figure_manifest.csv`. The complete input hashes and pinned toolkit commits are recorded in the adjacent `provenance.json`. The workflow is `scripts/article_figures/generate.py`; this report is built by `scripts/article_figures/build_report.py`.